

MS&E 125: Introduction to Applied Statistics
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Final Report: Analyzing the Evolution of Musical Features from 1950 to 2019

Research Question:

How have musical features like sadness and danceability changed from the 1950s (1950-1959) to today (2010-2019)? Are songs from different decades statistically different in these features?

Primary Dataset:

<https://www.kaggle.com/datasets/saurabhshahane/music-dataset-1950-to-2019>

This dataset provides lyrics and metadata for English-language songs released between 1950 and 2019. It includes features such as `artist_name`, `track_name`, `release_date`, `genre`, and audio-related attributes like `sadness`, `danceability`, `loudness`, `valence`, and `energy`. Additionally, the full lyrics for each track are included, offering insight into language and thematic content across decades. The dataset contains 23,689 song entries released between 1950 and 2019.

Note that all our outputs are attached further in the report.

Informative Plots

The first plot, titled *Average Sadness in Songs Over Time*, illustrates how the average sadness levels in popular music have evolved from the 1950s through the 2010s. The data show that songs in the 1950s and early 1960s tended to have higher sadness scores, with values peaking around 0.18. However, beginning in the late 1960s and especially after the 1970s, there is a noticeable downward trend. This decline becomes more pronounced in the 2000s, with the lowest levels of sadness appearing in the 2010s, falling below 0.10. This suggests that over the decades, the emotional tone of mainstream music has shifted away from somber or melancholic themes.

The second plot, *Average Danceability in Songs Over Time*, tracks the average danceability scores of songs across the same time span. In contrast to the sadness trend, danceability has steadily increased over time. In the late 1950s and 1960s, danceability scores hovered between 0.45 and 0.50, showing moderate variation. However, from the 1970s onward, the trend is consistently upward (though with some moderate stability), with significant increases especially evident after 2010. By the end of the 2010s, the average danceability surpasses 0.60, indicating that contemporary songs are much more rhythmically oriented and designed for physical engagement. This rise in danceability may reflect the growing popularity of genres such as disco, hip-hop, electronic, and pop music, which emphasize beats, groove, and bodily movement.

The third plot, *Average Sadness & Danceability Over Time*, presents both features on the same graph for direct comparison. It reveals a clear inverse trend: as danceability steadily increases across decades, average sadness declines. This suggests a broader shift in popular music toward more upbeat and rhythmically engaging songs, moving away from emotionally somber content.

Hypothesis Testing

We conducted a two-proportion z-test to evaluate whether the proportion of “sad” songs has changed significantly between the 1950s and the 2010s. While our original dataset contained Sadness scores that were continuous, we converted the data into binary outcomes by defining a song as “sad” if its sadness score exceeded 0.5. To conduct this test, we filtered the dataset to retain only those songs meeting this criterion and compared the proportions of sad songs in each decade. We chose significance level $\alpha = 0.05$.

H0: Sadness in music *has not* changed from the 1950s to the 2010s ($P_{2010s} = P_{1950s}$).

H1: Sadness in music *has* changed from the 1950s to the 2010s. ($P_{2010s} \neq P_{1950s}$).

$P_{2010s}^{\wedge}$: 0.0538 (calculated by dividing the number of sad songs released in the 2010s by the total number of songs released in the 2010s)

$P_{1950s}^{\wedge}$: 0.0339 (calculated using the process described above)

Type I Error: Incorrectly rejecting the null hypothesis when it is actually true. In this context, it would mean concluding that sadness in music has changed when, in fact, it has not.

Type II Error: Failing to reject the null hypothesis when it is actually false. In this case, it would mean concluding that sadness in music has not changed when it actually has.

Results: ($p = 0.00039 < 0.05$, $z\text{-statistic} = -3.549321$, $se = 0.0056$, 95% CI: [- 0.0324, -0.0074])

Statistical Conclusion:

In conclusion, since our p-value was less than $\alpha = 0.05$, we reject the null hypothesis and conclude that sadness in released songs *has* changed from the 1950s to the 2010s. Furthermore, we are 95% confident that the true difference in the proportion of sad songs between the 2010s and the 1950s lies between -3.24% and -0.74%. Since the entire interval is negative, this indicates that the proportion of sad songs in the 2010s is statistically significantly lower than in the 1950s. In other words, the results suggest that sadness in music has decreased from the 1950s to the 2010s, which aligns with what we see in the informative plots.

Regression Testing

To grow stronger in our hypothesis, we conducted regression testing by creating several regression model outputs, each measuring a different association between the variables in our

dataset. We produced four regression models — 1) sadness vs. time, 2) danceability vs. time, 3) sadness vs. danceability, and 4) sadness vs. danceability for each decade — which allowed for our team to draw a certain conclusion about the relationship between the two categorical variables.

In the first regression model, we predict **sadness vs. time**; the regression line indicates that sadness in songs is decreasing over time between 1950-2019.

In the second regression model, we predict **danceability vs. time**; the regression line indicates that danceability in songs is increasing over time between 1950-2019, contrasting the behavior of the sadness variable.

In the third regression model, we predict **sadness vs. danceability, with danceability as the independent variable**; the regression line indicates that there is a correlation between the two categorical variables with respect to the total index. Higher danceability in a song tends to be associated with lower sadness, on average.

In the fourth regression model, we plot **sadness vs. danceability for each decade from 1950-2010**. Because each regression line for each decade has a negative slope, we can draw the conclusion that higher danceability tends to be associated with lower sadness for each decade between 1950-2019.

The four different regression models all affirm our hypothesis that sadness in music has changed from the 1950s to the 2010s. Furthermore, measuring association for sadness with a variable as contrasting as danceability allowed us to more deeply analyze the context of our conclusions.

Drawbacks to Analysis

The first drawback of our regression analysis is that it is influenced by outliers and outside factors. A few extreme values (e.g., very high sadness scores in one year) can distort the trend line. Our data can be easily misinterpreted without acknowledgement of this context.

Additionally, music changes over time in part due to genre shifts (e.g., rise of EDM, hip hop). So just because sadness increases over time doesn't mean time causes sadness in music — it could be driven by other unmeasured factors like cultural shifts or production styles.

The dataset may also contain more songs in recent years due to better data collection or digital releases. This can bias the regression toward recent trends and distort long-term analysis.

Conclusion

Our report encapsulates the research we performed on two categorical variables of a very specific, complex, and fascinating dataset. For future research, we hope to be able to explore correlations between the other categorical variables like loudness, acousticness, etc. to reach stronger, more broad conclusions about the current trends in modern music. Further exploration of this dataset will empower us to answer some of our initial sample research questions:

- What musical features (like tempo or key) are most common in songs labeled as "happy" or "sad"? Can we predict a song's emotion label based on its audio features?
- Do solo, chamber, and orchestral music pieces differ in note density, instrument variety, or rhythm complexity? Can we find statistical differences across these groups?
- What audio features are most different between genres like jazz, classical, rock, and pop? Can we predict a song's genre using those features?
- Do popular songs have different audio features than less popular ones? Can we model popularity based on features like tempo, valence, or loudness?

For future research, we hope to...

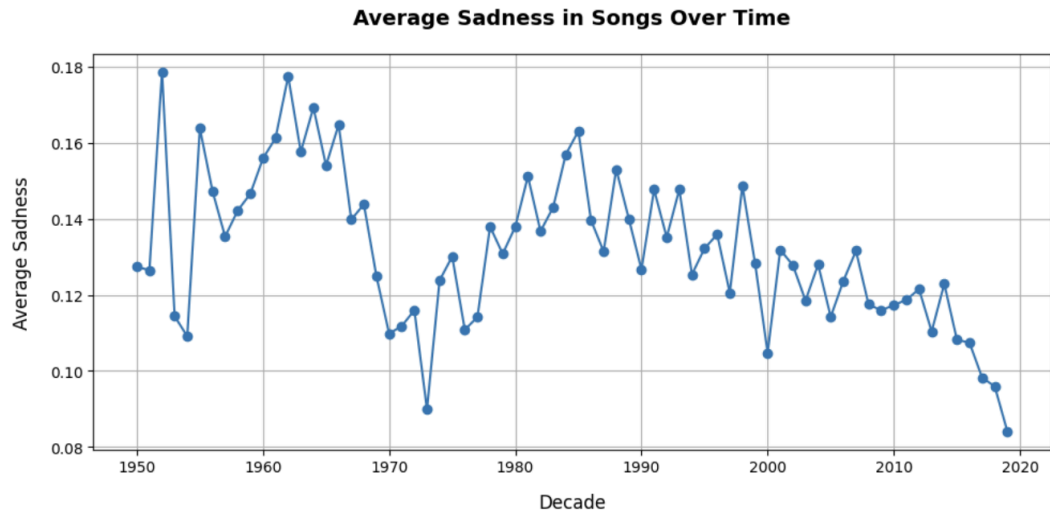
- Investigate how sadness and danceability vary across genres (e.g., pop, rock, hip hop) over time, which may reveal hidden shifts masked in the aggregate data.
- Extend the analysis to include variables like energy, valence, acousticness, or lyrical themes to better capture the emotional and structural complexity of music.
- Use multivariate regression or matching methods to account for possible confounders like genre, artist popularity, or lyrical language.

Thank you!

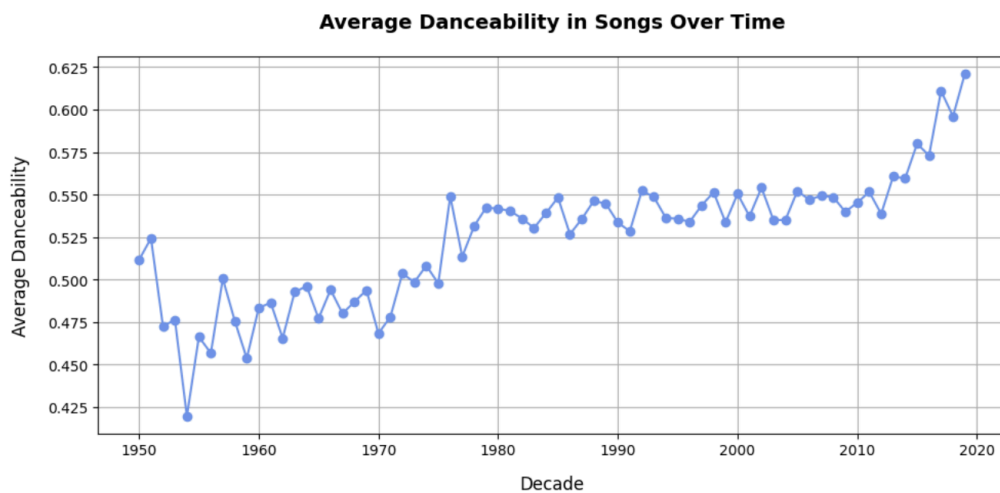
Appendix

Informative Plots:

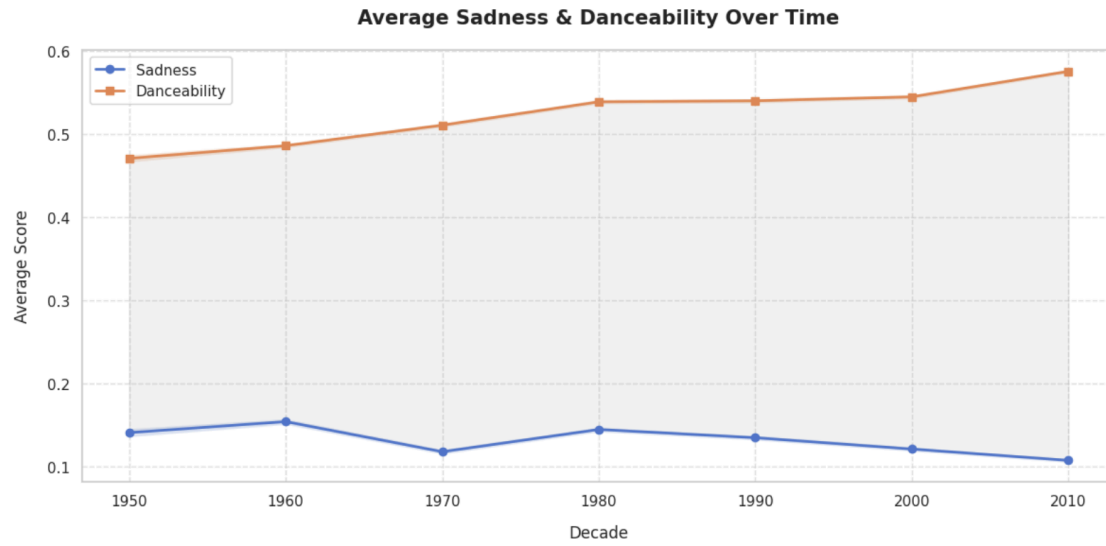
1) Average Sadness in Songs vs. Time



2) Average Danceability in Songs vs. Time

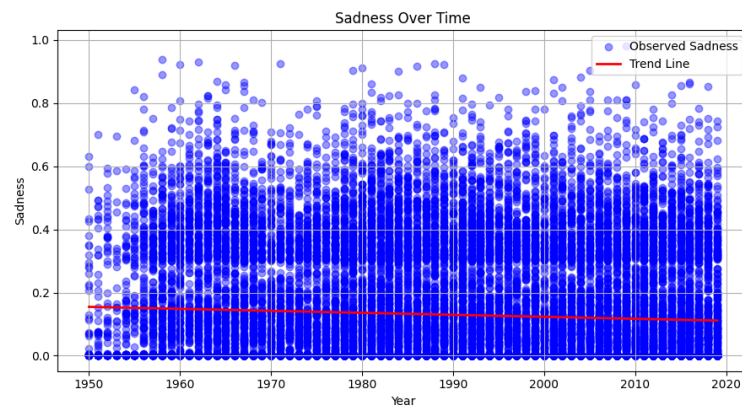


3) Average Sadness and Danceability vs. Time

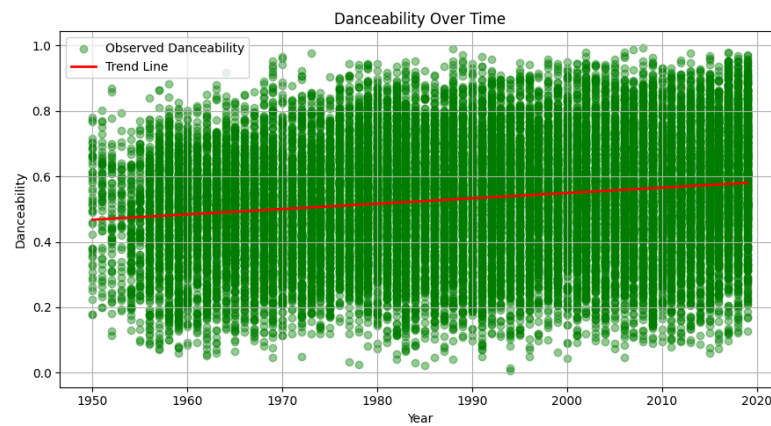


Regression Models:

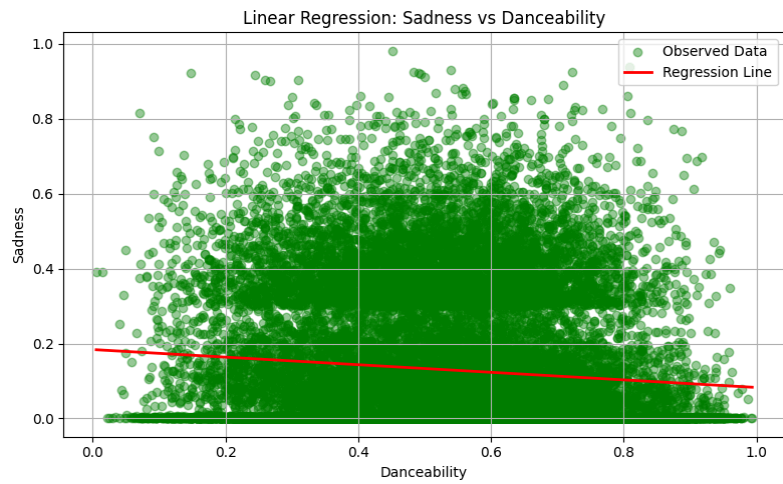
1) Sadness vs. Time



2) Danceability vs. Time



3) Sadness vs. Danceability



4) Sadness vs. Danceability For Each Decade

